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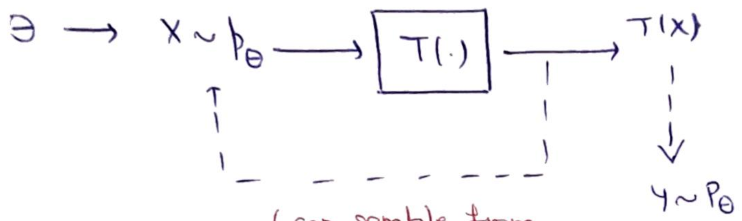

(can sample from P_θ without knowledge of θ).

ϕ How to find the sufficient statistics in a principled manner?

Theorem: (FACTORIZATION Theorem of fisher Neymen, Holmos - Savage, '40's)

let $X \sim P_\theta$, $\theta \in \Theta$ $T(X)$ is sufficient for θ iff -

$$P_\theta(x) = h(x)\, g_\theta(T(x))$$

$\forall x, \forall \theta$, where $h(x)$ does not depend on θ.

Note: The factorization may not be unique —

Proof: if part) Assume $\exists$ functions T, h, and g_θ such that $\forall \theta, x$: $P_\theta(x) = h(x)\, g_\theta(T(x))$

→ we will show that T is a sufficient statistics for θ.

we are showing for pmf.

$$P_\theta\left(X=x \mid T(X)=t\right)$$

If $T(x) \neq t$

$=0$

$$= \frac{P_\theta(X=x, T(X)=t)}{P_\theta[T(X)=t]} = \frac{P_\theta[X=x]}{P_\theta\{T(X)=t]}$$

$$= \frac{P_\theta(X=x)}{\sum_{y:T(y)=t} P_\theta[X=y]}$$

$$P_\theta(x) = h(x)\, g_\theta(T(x))$$

$$= \frac{h(x)\, g_\theta(t)}{\sum_{y:T(y)=t} h(y)\, g_\theta(t)}$$

$$= \frac{h(x)}{\sum_{y:T(y)=t} h(y)}$$

which doesnot depend on θ.

Only if) : Assume $T(X)$ is sufficient statistics for θ :

$\forall \theta, x$

$$P_\theta(x) = P_\theta(X=x)$$

$$= P_\theta(X=x, T(X)=T(x))$$

$$= P_\theta(T(X)=T(x))\,\underbrace{P_\theta(X=x \mid T(X)=T(x))}$$

$$= g_\theta(T(x)) \cdot h(x)$$

Cannot depend on θ, due to the suff. statistics assumptn

example : Sufficient stat. for Poi(λ) $\rightarrow$

$X \equiv (X_1 \dots X_n) \overset{iid}{\sim} \text{Poi}(\lambda)$

$$P_\lambda(\vec{x})^{(x_1 \dots x_n)} = \prod_{i=1}^{n} P_\lambda(x_i) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i}$$

$$= \underbrace{\left[e^{-n\lambda} \lambda^{\sum_{i=1}^{n} x_i} \right]}_{g_\lambda(x)} \underbrace{\left[\prod_{i=1}^{n} 1/x_i \right]}_{h(x)}$$

Hence, $\sum_{i=1}^{n} x_i$ is a sufficient satistics.

<u>example</u>: <u>Sufficient statistics for Uniform distribution</u> :-

$$(X_1 \dots X_n) \overset{iid}{\sim} \text{Unif } [0, \theta], \ \theta > 0$$

$$\forall x_1 \dots x_n \in \mathbb{R}^+, \ \forall \theta > 0$$

$$P_\theta(x_1 \dots x_n) = \prod_{i=1}^{n} P_\theta(x_i)$$

$$P_\theta(x_1 \dots x_n) = \prod_{i=1}^{n} \frac{1}{\theta} 1\{x_i \le \theta)$$

$$P_\theta(x_1 \dots x_n) = \left[\left(\frac{1}{\theta}\right)^n \prod_{i=1}^{n} 1\{x_i \le \theta\}\right] \cdot 1$$

$$P_\theta(x_1 \dots x_n) = \left[\left(\frac{1}{\theta}\right)^n 1\underbrace{\{\max_{i=1}^{n} x_i \le \theta\}}\right] \cdot 1$$

So $T(x) = \max_{i=1}^{n} x_i$ is a <u>sufficient statistics</u>.

Example: Sufficient statistics for <u>exponential family distributions</u>.

Ø An exponential family is a pdf or pmf having the form $\forall x, \theta$ $\boxed{P_\theta(x) = h(x)\, c(\theta) \exp\left[\sum_{i=1}^{k} w_i(\theta)\, t_i(x)\right]}$

$\forall x \in \mathcal{X}$
$\forall \theta \in \Theta$

where, → $h, t_1 \dots t_k$ are functions $(\mathcal{X} \to \mathbb{R})$ are not depending on θ.

→ $c, w_1 \dots w_k$ are functions $(\Theta \to \mathbb{R})$ not depending on θ.

real valued functions.

e.g Bernoulli –

$\forall x \{0, 1\}$ $\theta \in \{0, 1\}$

$P_\theta(x) = \theta^x (1-\theta)^{1-x}$

$P_\theta(x) = e^{[x \log\theta + (1-x)\log(1-\theta)]}$

hence $t_1(x) = x$ $\quad w_1(\theta) = \log\theta$

$t_2(x) = 1-x$ $\quad w_2(\theta) = \log(1-\theta)$

$h(x) = c(\theta) = 1$

Note:- $c(\theta)$ is the normalising Constant / "Partition function" of the distribution P_θ

e.g) Gaussian, Poisson, exponenial, pareto, Gamma ...

not e.g) – $\{\text{Unif}(0, \theta) : \theta > 0\}$

– Mixtures of Gaussian

e.g $Z \sim \text{Ber}(1/2)$

$X|_Z \sim N(z, \sigma^2)$

→ by the factorization theorem if $X = (X_1 \ldots X_n) \overset{iid}{\sim} P_\theta$ satisfying exponential family defn –

then, sufficient statistics for θ is –

here it is vector sufficient statistics

$$\left(\sum_{j=1}^{n} t_1(x_j) \quad \sum_{j=1}^{n} t_2(x_j) \quad \ldots \quad \sum_{j=1}^{n} t_k(x_j) \right)$$

eg $(X_1 \ldots X_n) \overset{iid}{\sim} Ber(\theta)$

$$\left(\sum_{j=1}^{n} x_j \, , \, \sum_{j=1}^{n} (1-x_j) \right) \longleftrightarrow \sum_{j=1}^{n} x_j$$

MINIMAL Sufficient : (statistics)

Ø A sufficient statistics is said to be minimal if, for any other sufficient statistics $T'(x)$, $T(x)$ is a function of $T'(x)$ ie

$\exists$ a function $f(\cdot)$ such that $T(x) = f(T'(x)) \; \forall x$

i.e if $x_1 \neq x_2$, $T'(x_1) = T'(x_2)$ then

$$T(x_1) = T(x_2)$$

Theorem : (test for minimal sufficient statistics)

let $X \sim P_\theta, \theta \in \Theta$. Suppose there exist a function $T(x)$ s.t

$\forall x, y \quad \frac{P_\theta(x)}{P_\theta(y)}$ does not depend on θ

iff $T(x) = T(y)$

Then, $T(X)$ is a minimal sufficient statistics

$\rightarrow$ (here reverse not hold)

Proof: Casella - Berger Thm. 6.2.13

eg Bernoulli (θ), $\theta \in [0,1]$

$X \equiv (X_1 \dots X_n) \sim Ber(\theta)$

$T(\mathbf{x}) = \sum_{i=1}^{n} x_i$

$\rightarrow$ take $x \equiv (x_1 \dots x_n) \in \{0,1\}^n$

$y \equiv (y_1 \cdots y_n) \in \{0,1\}^n$

$$\frac{P_\theta(x)}{P_\theta(y)} = \frac{(\theta)^{\sum x_i} (1-\theta)^{n-\sum x_i}}{(\theta)^{\sum y_i} (1-\theta)^{n-\sum y_i}}$$

$$= (\theta)^{(\sum x_i - \sum y_i)} (1-\theta)^{(\sum_i y_i - \sum_i x_i)}$$

$\equiv$ independent of θ iff $\sum x_i = \sum y_i$

i.e $T(x) = T(y)$

Hence $\sum_{i=1}^{n} x_i = T(x)$ is a minimal sufficient statistics